Abbring & Van den Berg (2003), built up step by step

Plan

1. One duration: survival, hazard, integrated hazard
2. People differ: unobserved heterogeneity and dynamic selection
3. The mixed proportional hazard (MPH) model
4. Two durations racing: competing risks
5. Adding a treatment: potential outcomes indexed by treatment time, no anticipation
6. What we actually observe
7. The model of the paper (Model 1a)
8. Identification, step by step
9. Extensions: heterogeneous effects, multiple spells
10. Connections: DiD with staggered adoption, sequential exogeneity, traps

Running example (the paper's, Sections 1 and 2.4)

- Unemployed workers receive unemployment benefits.
- At a random time S during the spell, their **benefits are reduced** (from level w to $b < w$).
- Outcome: unemployment duration Y.
- Question: what is the causal effect of the benefits reduction on the **transition rate to work**?

Road map: why each section is needed

- The effect is on a *rate* ⇒ we need the language of rates (Section 1).
- Workers differ in ways we do not see, and those still unemployed are selected ⇒ Sections 2–3.
- Before treatment, two things compete: finding a job vs. the benefits reduction ⇒ Section 4.
- Define the causal effect and what we observe ⇒ Sections 5–6.
- Put it together and show the effect is identified ⇒ Sections 7–8.

1 One duration

→ *Why now:* the paper's effect is on a *transition rate*. Before anything else: what is a rate for one duration, and how does it relate to probabilities?

Objects

- $Y \geq 0$: a continuous duration; y: a value.
- Density $f(y)$, CDF $F(y)$.
- Survival function $S(y) := \Pr(Y > y) = 1 - F(y)$: share still in the state at y.

1.1 Hazard rate

→ *Why now:* this is the object the treatment effect will act on.

$$\begin{aligned}
\theta(y) &:= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy \mid Y \ge y)}{dy} && \text{[definition]} \\
&= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy)}{dy \ \Pr(Y \ge y)} && \text{[conditional probability: } \Pr(A \mid B) = \Pr(A \cap B)/\Pr(B)\text{]} \\
&= \lim_{dy \downarrow 0} \frac{F(y + dy) - F(y)}{dy} \cdot \frac{1}{S(y)} && \text{[}Y \text{ continuous: } \Pr(Y \ge y) = \Pr(Y > y)\text{]} \\
&= \frac{f(y)}{S(y)} && \text{[definition of the derivative: } F' = f\text{]} \qquad (1)
\end{aligned}$$

- Exit rate *among those still in the state* at y.
- $\theta(y)\,dy \approx$ probability of leaving in $[y, y + dy)$, given still there at y.
- A rate, not a probability: can exceed 1.

1.2 Integrated hazard and survival

→ *Why now:* we model hazards, but data and likelihoods are about probabilities. We need the map from θ back to S and f; it is used in every later derivation.

$$\begin{aligned}
\theta(y) &= \frac{f(y)}{S(y)} && \text{[by (1)]} \\
&= -\frac{S'(y)}{S(y)} && \text{[}S = 1 - F\text{, so } S' = -f\text{]} \\
&= -\frac{d}{dy} \log S(y) && \text{[chain rule: } (\log S)' = S'/S\text{]}
\end{aligned}$$

$$\begin{aligned}
\Theta(y) := \int_0^y \theta(u)\,du &= -\big[\log S(u)\big]_0^y && \text{[fundamental theorem of calculus]} \\
&= -\log S(y) && \text{[}S(0) = 1\text{, so } \log S(0) = 0\text{]} \\
\implies \quad S(y) &= e^{-\Theta(y)} && \text{[exponentiate]} \qquad (2) \\
f(y) &= \theta(y)\, e^{-\Theta(y)} && \text{[}f = \theta S \text{ by (1)]} \qquad (3)
\end{aligned}$$

- Θ: integrated hazard.
- θ, Θ, S, f: four descriptions of the same distribution.

1.3 Deeper look: survival = surviving many small coin flips

→ *Why now:* (2) came out of calculus. Here is what it *means*, and it shows the key fact that every hazard is about a selected group.

Split $[0, y]$ into n intervals of length $\Delta = y/n$, starting at $u_1, \ldots, u_n$:

$$\begin{aligned}
S(y) &= \prod_{k=1}^{n} \Pr\big(\text{survive interval } k \mid \text{survived to } u_k\big) && \text{[multiplication rule, repeated]} \\
&= \prod_{k=1}^{n} \big(1 - \theta(u_k)\Delta + o(\Delta)\big) && \text{[definition of the hazard]} \\
\log S(y) &= \sum_{k=1}^{n} \log\big(1 - \theta(u_k)\Delta + o(\Delta)\big) && \text{[log of a product]} \\
&= -\sum_{k=1}^{n} \theta(u_k)\Delta + o(1) && \text{[}\log(1-z) = -z + O(z^2)\text{]} \\
&\xrightarrow{n\to\infty} -\int_0^y \theta(u)\, du = -\Theta(y) && \text{[Riemann sum]}
\end{aligned}$$

- Being still in the state at y = winning a sequence of coin flips, each with exit probability $\theta(u_k)\Delta$.
- Each coin flip is made *only by those who won all earlier ones.*
  - So a hazard always describes a **selected group**: the survivors.
  - With heterogeneous people this selection changes who is in the group (Section 2).
- Given the hazard, *when* the exit happens is pure chance (the coin flips). Section 7 exploits this for treatment timing.

1.4 Example: constant hazard

→ *Why now:* the simplest benchmark; it defines "duration dependence", which Section 2 will show is hard to read from data.

$$\begin{aligned}
&\theta(y) = c \quad \Longrightarrow \quad \Theta(y) = cy, \qquad S(y) = e^{-cy} && \text{[integrate; then (2)]} \\
&\Pr(Y > y + u \mid Y > y) = \frac{S(y+u)}{S(y)} && \text{[conditional probability; } \{Y > y+u\} \subset \{Y > y\}\text{]} \\
&\qquad\qquad = e^{-cu} && \text{[plug in } S\text{]}
\end{aligned}$$

- Does not depend on y: **memoryless**.
- θ decreasing in y: **negative duration dependence**. Increasing: **positive**.

Takeaway → next: with one homogeneous population, the hazard is directly read from data via $\theta = f/S$. Next: what if people differ in ways we do not observe?

2 Unobserved heterogeneity and dynamic selection

→ *Why now:* workers differ in unobserved ways (e.g. motivation). The paper compares treated and not-yet-treated workers *among those still unemployed.* So we must know what heterogeneity does to the group that is still there.

Model

- Individuals differ by an unobserved type $V \geq 0$ (motivation, skills, ...), fixed during the spell.
- G: distribution of V in the population, $G(v) = \Pr(V \leq v)$.

  - Notation: $dG(v)$ = "probability that V is (around) v": $\Pr(V = v)$ if discrete, $g(v)\,dv$ if continuous.
  - So $\int h(v)\,dG(v) = \mathbb{E}[h(V)]$.
- Section 1 applies to *each type*: $\theta(y \mid V) = f(y \mid V)/S(y \mid V)$.
- **Assumption:** this conditional hazard has the form

$$\theta(y \mid V) = \lambda(y)\,V.$$

  - $\lambda(y) = \theta(y \mid V = 1) = f(y \mid V = 1)/S(y \mid V = 1)$: the **baseline**, the hazard of the $V = 1$ type.
  - Common time profile for all; V scales its level.
  - "$V = 1$ type" needs a normalization, since (λ, V) and $(c\lambda, V/c)$ give the same hazard.
- Integrated baseline $\Lambda(y) := \int_0^y \lambda$. So $\Theta(y \mid V) = \Lambda(y)V$ and, by (2), $\Pr(Y > y \mid V) = e^{-\Lambda(y)V}$.
- The **population** hazard $\bar{\theta} = \bar{f}/\bar{S}$ (Section 1 applied to everyone pooled) is *not* assumed. It is derived below.

2.1 Population survival and hazard

→ *Why now:* we only observe the population, not each type. Compute the population hazard from the type hazards. Take discrete types v with shares $\Pr(V = v)$.

$$\begin{aligned}
\bar{S}(y) := \Pr(Y > y) &= \sum_v S(y \mid v)\Pr(V = v) && \text{[LIE (law of total probability)]} \\
\bar{f}(y) &= \sum_v f(y \mid v)\Pr(V = v) && \text{[$f = -S'$; differentiate the sum]} \\
\bar{\theta}(y) = \frac{\bar{f}(y)}{\bar{S}(y)} &= \sum_v \frac{f(y \mid v)\Pr(V = v)}{\bar{S}(y)} && \text{[(1) for the population]} \\
&= \sum_v \theta(y \mid v)\,\underbrace{\frac{S(y \mid v)\Pr(V = v)}{\bar{S}(y)}}_{=:\,w_y(v)} && \text{[$f(y \mid v) = \theta(y \mid v)S(y \mid v)$, by (1)]} \qquad (4)
\end{aligned}$$

- The population hazard is a **weighted average of the type hazards**.
- The weights $w_y(v)$ are *not* the population shares $\Pr(V = v)$: each share is multiplied by the type's survival $S(y \mid v)$.
- Weights sum to 1: $\sum_v S(y \mid v)\Pr(V = v) = \bar{S}(y)$.

2.2 What the weights are: the type mix among survivors

→ *Why now:* interpret the weights $w_y(v)$ in (4).

$$\begin{aligned}
\Pr(V = v \mid Y > y) &= \frac{\Pr(Y > y \mid V = v)\ \Pr(V = v)}{\Pr(Y > y)} && \text{[Bayes' rule]} \\
&= \frac{S(y \mid v)\ \Pr(V = v)}{\bar{S}(y)} = w_y(v) && \text{[definitions of $S(y \mid v)$, $\bar{S}$]} \qquad (5)
\end{aligned}$$

- $w_y(v)$ = probability that a person *still unemployed at* y is of type v = share of type v among survivors.

- Bayes flips the conditioning:
  - $S(y \mid v) = \Pr(\text{still there} \mid \text{type } v)$: given the type, chance to survive;
  - $w_y(v) = \Pr(\text{type } v \mid \text{still there})$: given survival, chance of the type.
- V is unobserved, so $w_y(v)$ cannot be counted in data; it follows from G and Λ once identified (Section 3).
- Example ($\lambda = 1$, types 0.3 and 1.5, half each, $y = 2$): $S(2 \mid v) \approx (0.55,\ 0.05)$, so $w_2 \approx (0.92,\ 0.08)$.

Now use the model $\theta(y \mid v) = \lambda(y)v$ and $S(y \mid v) = e^{-\Lambda(y)v}$ in (4):

$$\begin{aligned}
\bar{\theta}(y) &= \sum_v \lambda(y)\, v\ \Pr(V = v \mid Y > y) && \text{[(4) and (5)]} \\
&= \lambda(y) \sum_v v\ \Pr(V = v \mid Y > y) && \text{[linearity: } \lambda(y) \text{ does not depend on } v\text{]} \\
&= \lambda(y)\, \mathbb{E}[V \mid Y > y] && \text{[definition of a conditional mean]} \qquad (6) \\
\mathbb{E}[V \mid Y > y] &= \frac{\sum_v v\, e^{-\Lambda(y)v} \Pr(V = v)}{\sum_v e^{-\Lambda(y)v} \Pr(V = v)} = \frac{\mathbb{E}[V e^{-\Lambda(y)V}]}{\mathbb{E}[e^{-\Lambda(y)V}]} && \text{[plug in (5); definition of } \mathbb{E}\text{]} \qquad (7)
\end{aligned}$$

- Example: $\mathbb{E}[V \mid Y > 2] \approx 0.92 \cdot 0.3 + 0.08 \cdot 1.5 \approx 0.40$, versus $\mathbb{E}[V] = 0.9$. So $\bar{\theta}(2) \approx 0.40$ vs. $\bar{\theta}(0) = 0.9$.
- The observed hazard uses **survivors only**: $\lambda(y)\mathbb{E}[V \mid Y > y]$, not $\lambda(y)\mathbb{E}[V]$.

Note (continuous V). Replace $\Pr(V = v)$ by $dG(v)$ and sums by integrals. Then

$$dG(v \mid Y > y) = \frac{e^{-\Lambda(y)v}\, dG(v)}{\mathbb{E}[e^{-\Lambda(y)V}]},$$

and (6)–(7) are unchanged.

2.3 A name for the object that keeps appearing: the Laplace transform

→ *Why now:* every formula above contains $\mathbb{E}[e^{-aV}]$ for some a. Naming it and learning its properties is what makes all identification arguments below work.

Definition

$$\mathcal{L}(a) := \mathbb{E}\big[e^{-aV}\big] = \int_0^\infty e^{-av}\, dG(v), \qquad a \geq 0. \qquad (8)$$

- Reading in our model: $a = \Lambda(y)$ is the accumulated baseline hazard.
  - e^{-av}: probability that a type-v person is still unemployed.
  - $\mathcal{L}(a)$: share of the whole population still unemployed. Indeed $\bar{S}(y) = \mathcal{L}(\Lambda(y))$.
- Same family as the moment generating function: $\mathcal{L}(a) = M_V(-a)$, $M_V(u) := \mathbb{E}[e^{uV}]$.
  - Always exists, since $V \geq 0$ gives $0 < e^{-aV} \leq 1$.

Examples

Two types: v_1 with probability p, v_2 with probability $1 - p$:

$$\mathcal{L}(a) = p\, e^{-av_1} + (1 - p)\, e^{-av_2} \qquad \text{[expectation of a discrete r.v.]}$$

Gamma with mean 1 and variance $1/k$:

$$\begin{aligned}
\mathcal{L}(a) &= \int_0^\infty e^{-av}\,\frac{k^k v^{k-1} e^{-kv}}{\Gamma(k)}\,dv && \text{[definition (8)]}\\
&= \frac{k^k}{(k+a)^k}\int_0^\infty \frac{(k+a)^k v^{k-1} e^{-(k+a)v}}{\Gamma(k)}\,dv && \text{[multiply and divide by } (k+a)^k\text{]}\\
&= \Big(1+\frac{a}{k}\Big)^{-k} && \text{[integrand is a Gamma}(k, k+a)\text{ density]}
\end{aligned}$$

Properties

$$\begin{aligned}
\mathcal{L}(0) &= \mathbb{E}[1] = 1 && \text{[definition]}\\
\mathcal{L}'(a) &= -\mathbb{E}\big[Ve^{-aV}\big] < 0 && \text{[differentiate under } \mathbb{E}\text{; } <0 \text{ if } \Pr(V>0)>0\text{]}\\
\mathcal{L}^{(k)}(a) &= (-1)^k\,\mathbb{E}\big[V^k e^{-aV}\big] && \text{[repeat } k \text{ times]}\\
-\mathcal{L}'(0) &= \mathbb{E}[V] && \text{[set } a=0\text{: derivatives at 0 give moments]}\\
-\frac{\mathcal{L}'(a)}{\mathcal{L}(a)} &= \frac{\mathbb{E}[Ve^{-aV}]}{\mathbb{E}[e^{-aV}]} = \mathbb{E}[V \mid \text{survived to } a] && \text{[by (7)]} \qquad (9)\\
\implies \quad \bar\theta(y) &= \lambda(y)\Big(-\frac{\mathcal{L}'}{\mathcal{L}}\Big)\big(\Lambda(y)\big) && \text{[by (6)]}
\end{aligned}$$

Two facts we state without proof:

- **Uniqueness:** $\mathcal{L}$ determines G. Two distributions with the same $\mathcal{L}$ are equal.
- **Analyticity:** for $a > 0$, $\mathcal{L}$ equals its Taylor series.
  - So knowing $\mathcal{L}$ on *any small interval* pins it down on all of $(0, \infty)$.
  - Like a polynomial: a short stretch of its graph fixes the whole graph.

Why this matters for identification

- Survival data are always $\mathcal{L}$ evaluated at some index (here $\Lambda(y)$).
- Strategy throughout: learn the index; learn $\mathcal{L}$ on an interval; then analyticity + uniqueness give G.

2.4 The mean type among survivors falls

→ *Why now:* show the *direction* of the composition effect in (6).

$$\begin{aligned}
\mathbb{E}[V \mid Y>y] &= -\frac{d}{da}\log\mathcal{L}(a)\Big|_{a=\Lambda(y)} && \text{[(9); chain rule: } (\log\mathcal{L})' = \mathcal{L}'/\mathcal{L}\text{]}\\
\frac{d^2}{da^2}\log\mathcal{L}(a) &= \frac{\mathcal{L}''}{\mathcal{L}} - \Big(\frac{\mathcal{L}'}{\mathcal{L}}\Big)^2 && \text{[quotient rule]}\\
&= \mathbb{E}[V^2 \mid \cdot] - \mathbb{E}[V \mid \cdot]^2 = \mathrm{Var}(V \mid \text{survived to } a) && \text{[properties above: survivor moments]}\\
\frac{d}{dy}\mathbb{E}[V \mid Y>y] &= -\lambda(y)\ \mathrm{Var}(V \mid Y>y)\ \le\ 0 && \text{[chain rule: } \Lambda' = \lambda\text{]}
\end{aligned}$$

- The population hazard falls *even if no individual hazard falls.* This is **dynamic selection**.
- Hence two explanations for a falling exit rate:
  - true duration dependence: $\lambda\downarrow$ (e.g. skill loss: a *trap*);

– composition: $\mathbb{E}[V \mid Y > y] \downarrow$ (sorting on permanent heterogeneity).

- People still unemployed at month 6 are a selected group. Every comparison made at month 6 inherits this.

2.5 Deeper look: staying unemployed is evidence about your type

→ *Why now:* (5) is Bayes' rule. Reading it as learning about one person's type makes dynamic selection intuitive.

- Prior: $\Pr(V = v)$, the type mix at entry into unemployment (before we know anything about the person).
- Information: the person is still unemployed at y.
- Likelihood of that information given type v: $S(y \mid v)$. (A probability of an event, not a density.)
- Posterior $\propto$ prior $\times$ likelihood: exactly (5).
- The longer someone stays, the more the evidence points to a low v.

Gamma example (prior Gamma(k, k), mean 1, variance $1/k$):

$$\begin{aligned} dG(v \mid Y > y) &\propto v^{k-1} e^{-kv} \cdot e^{-\Lambda(y)v} && \text{[prior } \times \text{ likelihood]} \\ &= v^{k-1} e^{-(k+\Lambda(y))v} && \text{[collect exponents]} \\ \implies V \mid Y > y &\sim \text{Gamma}\big(k,\ k + \Lambda(y)\big) && \text{[recognize the kernel]} \\ \mathbb{E}[V \mid Y > y] &= \frac{k}{k + \Lambda(y)} = \frac{1}{1 + \Lambda(y)/k} && \text{[mean of Gamma}(k, r) \text{ is } k/r] \end{aligned} \tag{10}$$

- Falls from 1 toward 0 as $\Lambda(y)$ grows.
- Falls faster when $1/k$ (the variance of V) is larger: **more heterogeneity, more selection**.
- No heterogeneity ($k \to \infty$): $\mathbb{E}[V \mid Y > y] = 1$ for all y, so no selection.

2.6 Deeper look: two worlds that produce identical data

→ *Why now:* show that the confusion between "trap" and "sorting" is not just possible but *exact*. This is why Section 3 needs extra structure.

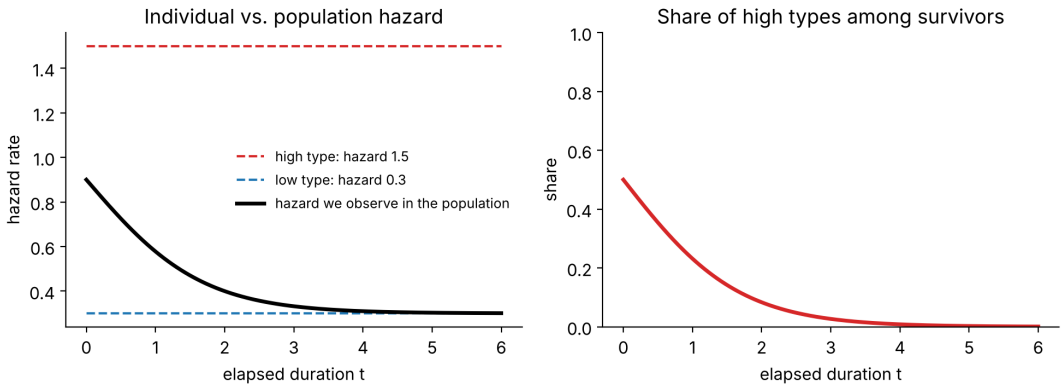
World H (heterogeneity, no duration dependence): $\lambda(y) = 1$, $V \sim \text{Gamma}(k, k)$.

$$\begin{aligned} \bar{S}_H(y) &= \mathcal{L}(\Lambda(y)) = \mathcal{L}(y) && [\bar{S} = \mathcal{L}(\Lambda);\ \Lambda(y) = y] \\ &= (1 + y/k)^{-k} && \text{[Gamma Laplace transform, Section 2.3]} \\ \bar{\theta}_H(y) &= -\frac{d}{dy} \log \bar{S}_H(y) = \frac{1}{1 + y/k} && [(1);\ \text{derivative of } k \log(1 + y/k)] \end{aligned}$$

World D (duration dependence, no heterogeneity): $V \equiv 1$, $\lambda_D(y) = \frac{1}{1+y/k}$.

$$\begin{aligned} \Theta_D(y) &= \int_0^y \frac{du}{1 + u/k} = k \log(1 + y/k) && \text{[antiderivative]} \\ S_D(y) &= e^{-k \log(1+y/k)} = (1 + y/k)^{-k} = \bar{S}_H(y) && [(2)] \end{aligned}$$

- World H: every individual has a *constant* hazard; the fall is pure sorting.
- World D: everyone is identical, and the fall is a true *trap*.
- Same distribution of Y. Data on one duration cannot tell them apart.



Two types with constant hazards 0.3 and 1.5 (equal shares). The population hazard falls from 0.9 to 0.3.

Takeaway → next: the observed hazard mixes λ (true duration dependence) and G (who is left). Next: can data separate them? Only with structure, which is the MPH model.

3 The mixed proportional hazard (MPH) model

→ *Why now:* Section 2 showed that λ and G are confounded in the observed hazard. Here: what structure lets data separate them? This is the one-duration version of the argument the paper runs with two durations (Section 4).

Model

- $\theta(y \mid x, V) = \lambda(y)\,\varphi(x)\,V$, with $V \sim G$ and $V \perp X$.
- Three multiplicative pieces: duration (λ), observed (φ), unobserved (V).
- Normalize $\varphi(x^*) = 1$ and $\Lambda(y^*) = 1$ for some chosen x^*, y^*.
  - Reason: (λ, φ, V) and $(c\lambda, \varphi, V/c)$ give the same hazard.

3.1 Survival through the Laplace transform

→ *Why now:* write what the data reveal, $S(y \mid x)$ and $\theta(y \mid x)$, in terms of the unknowns (λ, φ, G).

$$
\begin{aligned}
S(y \mid x) &= \mathbb{E}\big[e^{-\varphi(x)\Lambda(y)V}\big] && \text{[LIE over } V\text{; (2); } V \perp X\text{]} \\
&= \mathcal{L}\big(\varphi(x)\Lambda(y)\big) && \text{[definition (8)]} \qquad (11) \\
\theta(y \mid x) &= \lambda(y)\varphi(x)\Big(-\frac{\mathcal{L}'}{\mathcal{L}}\Big)\big(\varphi(x)\Lambda(y)\big) && \text{[}-\tfrac{d}{dy}\log \text{ of (11); chain rule]} \qquad (12)
\end{aligned}
$$

- Data reveal one function $\mathcal{L}$ of a **single index** $\varphi(x)\Lambda(y)$.
- x and y both move the index: this is what the three steps below exploit.

3.2 Deeper look: why covariates reveal heterogeneity

→ *Why now:* Section 2.6 showed one duration is not enough. Covariates break the tie, but *how*? Answer: heterogeneity leaves a fingerprint on how covariate effects evolve with duration.

Take the two worlds of Section 2.6 and add a covariate with $\varphi(x)$. Compare groups x_1, x_2 with $\varphi_1 > \varphi_2$.

World D (no heterogeneity): $\theta(y \mid x) = \lambda_D(y)\varphi(x)$, so

$$\frac{\theta(y \mid x_1)}{\theta(y \mid x_2)} = \frac{\varphi_1}{\varphi_2} \quad \text{for all } y. \qquad [\lambda_D(y) \text{ cancels}]$$

World H (Gamma heterogeneity): $-\mathcal{L}'(a)/\mathcal{L}(a) = \frac{d}{da}\, k \log(1 + a/k) = \frac{1}{1+a/k}$, so

$$\begin{aligned} \theta(y \mid x) &= \lambda(y)\, \varphi(x)\, \frac{1}{1 + \varphi(x)\Lambda(y)/k} && [(12)] \\ \frac{\theta(y \mid x_1)}{\theta(y \mid x_2)} &= \frac{\varphi_1}{\varphi_2} \cdot \frac{1 + \varphi_2\Lambda(y)/k}{1 + \varphi_1\Lambda(y)/k} && [\text{divide; } \lambda(y) \text{ cancels}] \\ &\xrightarrow{y\downarrow 0} \frac{\varphi_1}{\varphi_2}, \qquad \xrightarrow{y\to\infty} 1 && [\Lambda(0) = 0;\ \Lambda \to \infty] \end{aligned}$$

- With heterogeneity, the covariate effect **fades** with duration.
  - The high-φ group loses its high-V members faster, so its survivors become more negatively selected.
- Without heterogeneity it stays constant.
- So the MPH assumption reads: *absent selection, covariate effects would be constant over duration.*
  - Any change in the ratio over duration is then attributed to selection, which identifies G.
- Step 1 below reads the "true" ratio at $y \approx 0$, before selection has acted.
- Caveat: if true covariate effects vary with duration (non-proportional), this logic breaks.

3.3 Identification (Elbers & Ridder 1982) in three steps

→ *Why now:* recover φ, then G, then λ from $S(y \mid x)$. Each step uses one property of $\mathcal{L}$ from Section 2.3. Section 4 repeats these steps in two dimensions.

Step 1: φ from short durations. Assume $\mathbb{E}V < \infty$.

$$\begin{aligned} \frac{\theta(y \mid x)}{\theta(y \mid x^*)} &= \varphi(x)\, \frac{(-\mathcal{L}'/\mathcal{L})(\varphi(x)\Lambda(y))}{(-\mathcal{L}'/\mathcal{L})(\Lambda(y))} && [\text{by (12); } \lambda(y) \text{ cancels; } \varphi(x^*) = 1] \\ &\xrightarrow{y\downarrow 0} \varphi(x)\, \frac{-\mathcal{L}'(0)/\mathcal{L}(0)}{-\mathcal{L}'(0)/\mathcal{L}(0)} = \varphi(x) && [\Lambda(y) \to 0;\ \text{continuity; } -\mathcal{L}'(0) = \mathbb{E}V < \infty] \end{aligned}$$

- At $y \approx 0$ nobody has been selected out yet, so the ratio is free of selection.
- Uses short durations, where data are thin.

Step 2: G at the normalization date.

$$S(y^* \mid x) = \mathcal{L}\big(\varphi(x)\big) \qquad [(11) \text{ with } \Lambda(y^*) = 1]$$

- $\varphi(x)$ known from step 1. Varying x traces $\mathcal{L}$ on an interval.
- Analytic ⇒ $\mathcal{L}$ known on $(0, \infty)$ ⇒ G known.

Step 3: baseline.

$$\Lambda(y) = \frac{\mathcal{L}^{-1}\big(S(y \mid x)\big)}{\varphi(x)} \qquad [\text{invert (11); } \mathcal{L} \text{ strictly decreasing}]$$

What was assumed: proportionality, $V \perp X$, $\mathbb{E}V < \infty$, variation in $\varphi(x)$.

Takeaway → next: with MPH structure and covariates, one duration reveals λ, φ and the full distribution of unobserved types. In the paper, *before* treatment, two durations compete: job vs. benefits reduction. Their link through unobserved types is exactly the selection problem. Next: extend the argument to two durations.

4 Two durations racing: competing risks

→ *Why now:* before any treatment, a worker either finds a job (Y) or has benefits reduced (S), whichever comes first. If the same unobserved traits drive both (e.g. low motivation ⇒ slow job finding *and* early reduction), the treated are selected. This section asks: can the dependence between the two be learned from the race alone? No treatment effect is involved yet.

Set-up

- Two latent durations: Y (exit to work) and S (benefits reduction).
- Only the **first** event is seen, plus which one it was.
- MPH for each, with **correlated** unobservables $V = (V_Y, V_S) \sim G$, $V \perp X$:

$$\theta_Y(t \mid x, V) = \lambda_Y(t)\varphi_Y(x)V_Y, \qquad \theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S.$$

- Given (x, V): $Y \perp S$. All dependence runs through V.
- Bivariate Laplace transform: $\mathcal{L}(a, b) := \mathbb{E}[e^{-aV_Y - bV_S}]$; partials $\mathcal{L}_a, \mathcal{L}_b, \mathcal{L}_{ab}$.
  - Same properties as Section 2.3: derivatives give moments among survivors; uniqueness; analyticity.
- Indices: $a_t(x) := \varphi_Y(x)\Lambda_Y(t)$, $b_t(x) := \varphi_S(x)\Lambda_S(t)$.

4.1 Joint survival

→ *Why now:* the model's implication for the two latent durations together.

$$\begin{aligned}
\Pr(Y > y, S > s \mid x) &= \mathbb{E}\big[\Pr(Y > y, S > s \mid x, V)\big] && \text{[LIE over } V\text{; } V \perp X\text{]} \\
&= \mathbb{E}\big[\Pr(Y > y \mid x, V)\Pr(S > s \mid x, V)\big] && \text{[} Y \perp S \mid (x, V)\text{]} \\
&= \mathbb{E}\big[e^{-\varphi_Y(x)\Lambda_Y(y)V_Y}\, e^{-\varphi_S(x)\Lambda_S(s)V_S}\big] && \text{[(2) for each]} \\
&= \mathcal{L}\big(\varphi_Y(x)\Lambda_Y(y),\, \varphi_S(x)\Lambda_S(s)\big) && \text{[definition of } \mathcal{L}\text{]} \qquad (13)
\end{aligned}$$

4.2 What the data reveal

→ *Why now:* we never see both durations, only the first event. Derive which pieces of (13) are observable.

Probability that nothing has happened by t (the diagonal $y = s = t$):

$$P_0(t \mid x) := \Pr(Y > t, S > t \mid x) = \mathcal{L}\big(a_t(x), b_t(x)\big) \qquad \text{[(13) at } y = s = t\text{]} \qquad (14)$$

Cause-specific hazard of exit (rate of Y happening first, among those with nothing yet):

$$\begin{aligned}
h_Y(t \mid x) &:= \lim_{dt \downarrow 0} \frac{\Pr(t \le Y < t + dt,\ S > t \mid x)}{dt\ P_0(t \mid x)} && \text{[definition]} \\
&= \frac{-\partial_y \Pr(Y > y, S > t \mid x)\big|_{y=t}}{P_0(t \mid x)} && \text{[derivative of a survival in } y\text{]} \\
&= \frac{-\mathcal{L}_a(a_t, b_t)\, \varphi_Y(x) \lambda_Y(t)}{\mathcal{L}(a_t, b_t)} && \text{[chain rule on (13); } \Lambda_Y' = \lambda_Y\text{]} \\
&= \lambda_Y(t) \varphi_Y(x)\, \mathbb{E}[V_Y \mid R_t, x] && [-\mathcal{L}_a/\mathcal{L} = \text{survivor mean, as in (7)}] \qquad (15)
\end{aligned}$$

where $R_t := \{Y > t, S > t\}$ is the **risk set**. Symmetrically, $h_S(t \mid x) = \lambda_S(t)\varphi_S(x)\mathbb{E}[V_S \mid R_t, x]$.

4.3 Where the dependence of V_Y, V_S lives

→ *Why now:* the selection is the dependence of V_Y and V_S. Locate it inside the observable P_0.

Write $w(v) \propto e^{-av_Y - bv_S}$ (survivor weights) and $\mathbb{E}_w$ for expectations under them.

$$\begin{aligned}
\partial_a \log \mathcal{L} &= \frac{\mathcal{L}_a}{\mathcal{L}} = -\mathbb{E}_w[V_Y] && \text{[chain rule; differentiate under } \mathbb{E}\text{]} \\
\partial_b \partial_a \log \mathcal{L} &= \frac{\mathcal{L}_{ab}}{\mathcal{L}} - \frac{\mathcal{L}_a \mathcal{L}_b}{\mathcal{L}^2} && \text{[quotient rule]} \\
&= \mathbb{E}_w[V_Y V_S] - \mathbb{E}_w[V_Y]\mathbb{E}_w[V_S] && [\mathcal{L}_{ab} = \mathbb{E}[V_Y V_S e^{\cdots}],\ \mathcal{L}_b/\mathcal{L} = -\mathbb{E}_w[V_S]] \\
&= \mathrm{Cov}(V_Y, V_S \mid R_t, x) && \text{[definition of covariance; at } (a,b) = (a_t, b_t)\text{]} \qquad (16)
\end{aligned}$$

- $V_Y \perp V_S \iff \mathcal{L}(a,b) = \mathcal{L}_Y(a)\mathcal{L}_S(b) \iff \log P_0$ additively separable in the two indices.
- Dependence (selection) = an **interaction** of the two indices in $\log P_0$.

4.4 Deeper look: the fingerprint of selection in the race

→ *Why now:* (16) is abstract. Translate it into something observable: how a shift in treatment intensity changes the exit rate of those *not* treated.

Hold the exit index a fixed and raise the treatment index b (a group where reductions happen more often):

$$\begin{aligned}
h_Y &= \lambda_Y \varphi_Y\, \mathbb{E}_w[V_Y], \qquad w \propto e^{-aV_Y - bV_S} && [(15)] \\
\frac{\partial}{\partial b}\mathbb{E}_w[V_Y] &= -\mathbb{E}_w[V_Y V_S] + \mathbb{E}_w[V_Y]\mathbb{E}_w[V_S] && \text{[quotient rule, as in Section 2.4]} \\
&= -\,\mathrm{Cov}(V_Y, V_S \mid R_t, x) && \text{[definition of covariance]} \\
\implies \quad \frac{\partial \log h_Y}{\partial b} &= -\frac{\mathrm{Cov}(V_Y, V_S \mid R_t, x)}{\mathbb{E}[V_Y \mid R_t, x]} && \text{[log derivative; } \lambda_Y\varphi_Y \text{ do not depend on } b\text{]} \qquad (17)
\end{aligned}$$

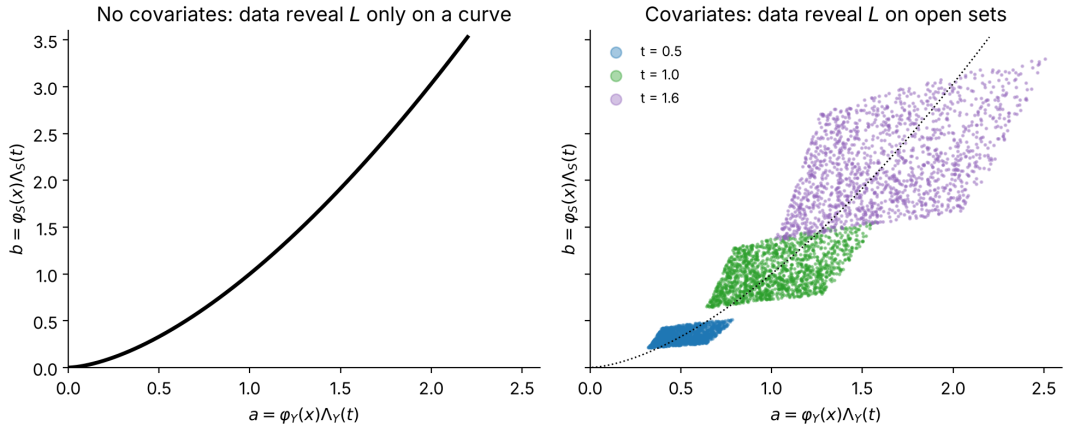
- Those in the risk set are *untreated*. By no anticipation, treatment intensity has no causal effect on their exit rate.
- So any response of their exit rate to b is **pure selection**:
  - more treatment removes high-V_S people from the untreated pool;
  - if high V_S goes with high V_Y, the untreated survivors are left with lower V_Y, so their exit rate falls.
- No selection (Cov = 0): shifting treatment intensity leaves the untreated exit rate unchanged.

- This is the IV-like role of covariates:
  - no variable needs to be excluded from the exit hazard;
  - but we must move b *holding a fixed*, i.e. x must shift the two indices in different directions (A3).
- Treatment acts like a competing exit: it removes people from the untreated group, just as job-finding removes people in Section 2.

4.5 The problem: Tsiatis (1975)

→ *Why now:* the dependence sits in $\mathcal{L}$, but do the data reveal enough of $\mathcal{L}$? Without structure, no.

- Without structure, any competing-risks data can be generated by *independent* Y, S.
- Geometry, without covariates:
  - the data reveal $\mathcal{L}$ only at points $(\Lambda_Y(t), \Lambda_S(t))$, i.e. along a **curve**;
  - a curve cannot pin down a function of two variables, so the dependence is not identified.
- With covariates moving φ_Y and φ_S separately: points $(a_t(x), b_t(x))$ fill **open sets**.



4.6 Identification (Prop. 2) in three steps

→ *Why now:* show that MPH + covariates fix the Tsiatis problem: the steps of Section 3.3, in two dimensions.

Assumptions. (A3) $\{(\varphi_Y(x), \varphi_S(x))\}$ contains an open set of $\mathbb{R}^2$. (A4) $\mathbb{E}V_Y, \mathbb{E}V_S < \infty$. Normalize $\varphi_Y(x^*) = \varphi_S(x^*) = 1$, $\Lambda_Y(t^*) = \Lambda_S(t^*) = 1$.

Step 1: φ_Y, φ_S**.**

$$\frac{h_Y(t \mid x)}{h_Y(t \mid x^*)} = \varphi_Y(x)\,\frac{\mathbb{E}[V_Y \mid R_t, x]}{\mathbb{E}[V_Y \mid R_t, x^*]} \qquad \text{[by (15); } \lambda_Y(t) \text{ cancels]}$$

$$\xrightarrow{t\downarrow 0} \varphi_Y(x)\,\frac{\mathbb{E}V_Y}{\mathbb{E}V_Y} = \varphi_Y(x) \qquad \text{[risk set} \to \text{population; } V \perp X\text{; (A4)]}$$

Same for φ_S with h_S. Uses *which* event happened.

Step 2: G**, including the dependence.**

$$P_0(t^* \mid x) = \mathcal{L}\big(\varphi_Y(x), \varphi_S(x)\big) \qquad \text{[(14) with } \Lambda_Y(t^*) = \Lambda_S(t^*) = 1\text{]}$$

- Known points fill an open set (A3) ⇒ $\mathcal{L}$ known there.

- Analytic $\Rightarrow$ known on $(0,\infty)^2 \Rightarrow$ joint G known.

Step 3: baselines. For each t, $(\Lambda_Y(t), \Lambda_S(t))$ is the unique (a,b) with

$$P_0(t \mid x) = \mathcal{L}\big(\varphi_Y(x)a,\ \varphi_S(x)b\big) \quad \text{for all } x. \qquad \text{[(14); } \mathcal{L}, \varphi\text{'s known]}$$

- Uniqueness: two solutions would give $\mathcal{L}(u,v) = \mathcal{L}(ku, mv)$ on $(0,\infty)^2$, $(k,m) \neq (1,1)$.
- Iterating, $\mathcal{L}$ would be constant, contradicting strict monotonicity.
- These three steps are our reconstruction; the paper cites Abbring & Van den Berg (2003, JRSSB).

Takeaway $\rightarrow$ next: the race alone reveals $\lambda_Y, \lambda_S, \varphi_Y, \varphi_S$ and the joint G, i.e. how selected those still at risk are. Next: introduce the treatment and define its causal effect.

5 Adding a treatment

$\rightarrow$ *Why now:* so far no causal effect. We need to define "the effect of a benefits reduction at s", and say when the pre-treatment race of Section 4 is untouched by the treatment.

5.1 Potential outcomes indexed by treatment time

$\rightarrow$ *Why now:* a dynamic treatment is a choice of *date*, so the causal objects are one outcome per date.

- Treatment = benefits reduction at date $s \in [0,\infty]$; $s = \infty$: never.
- One potential outcome per date: $Y^*(s)$, with hazard $\theta_{Y^*(s)}$, integrated hazard $\Theta_{Y^*(s)}$.
- Each $Y^*(s)$ is fully described by its hazard, so "the effect of s" = how the hazard changes with s.
- Actual treatment date S; observed outcome $Y = Y^*(S)$.

5.2 No anticipation (Assumption 1)

$\rightarrow$ *Why now:* for Section 4 to apply before treatment, the future treatment date must not affect the hazard before it.

$$\Theta_{Y^*(s)}(y) = \Theta_{Y^*(s')}(y) \qquad \text{for all } y \leq \min\{s, s'\}.$$

Take $s' = \infty$ and differentiate in y:

$$\theta_{Y^*(s)}(y) = \theta_{Y^*(\infty)}(y) \qquad \text{for } y < s. \qquad \text{[derivative of both sides]}$$

- Before treatment, everyone has the **untreated** hazard, whatever their future date.
- Economic content: people do not act on knowledge of their own future reduction date.
- Violated if, e.g., the date is announced in advance and people search harder before it.
- Not ruled out: reacting to the *policy* (e.g. the rate at which reductions happen).
  - This reaction is in every $Y^*(s)$.
  - So the effect is "treated at s vs. later, under this policy", not "policy vs. no policy".
- Prop. 1: without more structure, data cannot distinguish (i) causal effects from selection, nor (ii) anticipation from no anticipation. Hence Assumption 1 is maintained, and selection must be modelled.

Takeaway $\rightarrow$ next: under no anticipation, the period before treatment is a pure race, as in Section 4. Next: what exactly do the data contain?

6 What we actually observe

→ *Why now:* identification means recovering the model from the data distribution. We need that distribution precisely, including what is censored.

The censoring

- S is seen only if $S < Y$: if the worker finds a job first, the reduction date is never seen.
- Paper's data (per x): $Q_S(y,s) := \Pr(Y > y, S > s, Y > S)$ and $Q_Y(y) := \Pr(Y > y, Y < S)$. Assume $\Pr(Y = S) = 0$.

Pre-treatment objects

$$\begin{aligned}
P_0(t) &= \Pr(Y > t, S > t, Y > S) + \Pr(Y > t, S > t, Y < S) && \text{[partition by } Y \gtrless S;\ \Pr(Y=S)=0\text{]}\\
&= Q_S(t,t) + Q_Y(t) && \text{[}Y > t, Y < S \text{ implies } S > t\text{]}\\
h_Y(t) &= \frac{-Q_Y'(t)}{P_0(t)} && \text{[}-Q_Y'\text{: density of untreated exits]}\\
h_S(t) &= \frac{-\partial_s Q_S(y,s)\big|_{y\downarrow t,\, s=t}}{P_0(t)} && \text{[density of } S \text{ at } t\text{, still unemployed]}
\end{aligned}$$

Post-treatment object: survival of the cohort treated at s. For $y > s$:

$$\begin{aligned}
-\partial_s Q_S(y,s) &= \lim_{ds\downarrow 0} \frac{\Pr(Y > y,\ s \le S < s + ds,\ Y > S)}{ds} && \text{[derivative of } Q_S \text{ in } s\text{]}\\
&= \lim_{ds\downarrow 0} \frac{\Pr(Y > y,\ s \le S < s + ds)}{ds} && \text{[}Y > y > S \text{ implies } Y > S\text{]}\\
P_1(y \mid s) &:= \Pr(Y > y \mid S = s,\, Y > s) && \text{[definition]}\\
&= \frac{\partial_s Q_S(y,s)}{\lim_{y'\downarrow s} \partial_s Q_S(y',s)} && \text{[cond. probability; } ds \text{ cancels]} \qquad (18)
\end{aligned}$$

- (P_0, h_Y, h_S) = competing-risks data, as in Section 4.
- P_1 = the only place where the treatment effect can show up.

Takeaway → next: pre-treatment data identify the race (Section 4). Post-treatment data P_1 must deliver δ. Next: the model linking both.

7 The model of the paper (Model 1a)

→ *Why now:* combine the pieces: the competing-risks MPH model of Section 4 before treatment, and a causal effect δ after it.

Assumption 2 (selection on observables and unobservables): $Y^*(\cdot) \perp S \mid (X, V)$, with V time-invariant.

Model 1a

$$\theta_Y(t \mid s, x, V) = \begin{cases} \lambda_Y(t)\varphi_Y(x)V_Y & t \le s \\ \lambda_Y(t)\varphi_Y(x)V_Y\, \delta(t \mid s, x) & t > s \end{cases}$$
$$\theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S, \qquad (V_Y, V_S) \sim G, \quad V \perp X.$$

- δ: **the treatment effect**, a multiplier on the exit rate after treatment.

  - May depend on t, s (hence on time since treatment $t-s$), and x; not on V.
- No anticipation is built in: δ only acts for $t > s$.
- Selection on unobservables: V_Y and V_S may be dependent in any way.
- The same x in both hazards: **no exclusion restriction**.
- $\lambda_Y, \lambda_S, \varphi_Y, \varphi_S, G, \delta$ all nonparametric.
- Plus (A3), (A4) from Section 4.

Integrated outcome hazard, with $D(y \mid s, x) := \int_s^y \lambda_Y(\tau)\delta(\tau \mid s, x)\, d\tau$:

$$\begin{aligned}\Theta_Y(y \mid s, x, V) &= \int_0^s \lambda_Y \varphi_Y V_Y \, d\tau + \int_s^y \lambda_Y \varphi_Y V_Y \delta \, d\tau && \text{[split the integral at } s\text{]} \\ &= \varphi_Y(x) V_Y \big[\Lambda_Y(s) + D(y \mid s, x)\big] \quad (y > s) && \text{[linearity of the integral]} \end{aligned} \tag{19}$$

Which part of the treatment time is pure chance?

→ *Why now:* for causal inference we need variation in treatment that is unrelated to the outcome. In IV it comes from an excluded variable; here there is none. So: given (x, V), is anything left in S that is pure luck?

Trick: transform S by its own integrated hazard, $E_S := \Theta_S(S \mid x, V) = \varphi_S(x) V_S \Lambda_S(S)$. Fix (x, V); for $e \geq 0$:

$$\begin{aligned}\Pr(E_S > e \mid x, V) &= \Pr\big(\Theta_S(S \mid x, V) > e \mid x, V\big) && \text{[definition of } E_S\text{]} \\ &= \Pr\big(S > \Theta_S^{-1}(e \mid x, V) \mid x, V\big) && \text{[}\Theta_S \text{ strictly increasing in } S\text{]} \\ &= \exp\big(-\Theta_S(\Theta_S^{-1}(e \mid x, V) \mid x, V)\big) && \text{[(2), conditional on } (x, V)\text{]} \\ &= e^{-e} && \text{[}\Theta_S(\Theta_S^{-1}(e)) = e\text{]}\end{aligned}$$

- The right side does not depend on (x, V). So $E_S \sim \text{Exp}(1)$ and $E_S \perp (x, V)$.
- (Same as $F(S) \sim U(0,1)$ for any continuous S: indeed $E_S = -\log \Pr(S > s \mid x, V)\big|_{s=S}$.)

Invert the definition of E_S:

$$S = \Lambda_S^{-1}\Big(\frac{E_S}{\varphi_S(x)\, V_S}\Big) \qquad \text{[solve } \varphi_S V_S \Lambda_S(S) = E_S \text{ for } S\text{]}$$

- The treatment time splits into a **systematic part** (x, V_S) and **pure luck** E_S.
- V_S is the source of selection (it may be correlated with V_Y).
- E_S moves S around, and by Assumption 2 it does not enter Y's hazard.
  - It plays the role of an instrument. But what is excluded is an unobserved **shock**, not an observed variable.
  - This is the paper's footnote 15: in each instant, treatment is a coin flip with probability $\theta_S\, dt$ (Section 1.3).
- Contrast with a static threshold model $D = \mathbf{1}\{\mu(x) > U\}$: given (x, U), D is deterministic, so no luck is left.

Takeaway → next: the model has a random component in timing (like an instrument) and selection through V (to be learned from the race). Next: show δ is identified.

8 Identification, step by step

→ *Why now:* everything is in place. Plan:

- A: learn the selection (G) from pre-treatment data;
- B: describe who is treated at s;
- C: the simplest case, the moment of treatment;
- D: the general case.

8.1 Step A: the pre-treatment data are the competing-risks model of Section 4

→ *Why now:* check that the pre-treatment data are exactly the race of Section 4, so its results apply.

$$\begin{aligned} P_0(t \mid x) &= \mathbb{E}\big[\Pr(Y > t, S > t \mid x, V)\big] && \text{[LIE; } V \perp X\text{]} \\ &= \mathbb{E}\big[e^{-\varphi_Y(x)\Lambda_Y(t)V_Y}\, e^{-\varphi_S(x)\Lambda_S(t)V_S}\big] && \text{[Assumption 2; before } S\text{, } \delta \text{ absent (no anticipation)]} \\ &= \mathcal{L}\big(a_t(x), b_t(x)\big) && \text{[as in (14)]} \end{aligned}$$

- Same for h_Y, h_S. So Prop. 2 (Section 4.6) applies: $\varphi_Y, \varphi_S, \Lambda_Y, \Lambda_S, G$ are identified.
- This uses no post-treatment data.

8.2 Step B: who gets treated at s

→ *Why now:* the effect is estimated by following the cohort treated at s. We need its type distribution.

Risk set at s:

$$\begin{aligned} dG_{R_s}(v) &= \frac{\Pr(R_s \mid x, v)\, dG(v)}{\Pr(R_s \mid x)} && \text{[Bayes' rule]} \\ &\propto e^{-a_s v_Y - b_s v_S}\, dG(v) && \text{[as in Step A]} \end{aligned}$$

Cohort treated at s (in the risk set and treated in $[s, s+ds)$):

$$\begin{aligned} dG_s(v) &\propto \Pr(s \le S < s + ds \mid R_s, x, v)\, dG_{R_s}(v) && \text{[Bayes' rule]} \\ &\propto \lambda_S(s)\varphi_S(x)v_S\, ds\, dG_{R_s}(v) && \text{[definition of the hazard } \theta_S\text{]} \\ &\propto v_S\, e^{-a_s v_Y - b_s v_S}\, dG(v) && \text{[drop factors not depending on } v\text{]} \end{aligned} \tag{20}$$

- Treated cohort = risk set **reweighted by the treatment intensity** v_S.
- Fully known after Step A.

8.3 Step C: the selection problem at the moment of treatment

→ *Why now:* the cleanest comparison: right at s, compare the just-treated with those not yet treated. This isolates the selection problem in one factor.

Exit hazard of the just-treated vs. the not-yet-treated, at s:

$$\begin{aligned} \text{just treated:} \quad & \lambda_Y(s)\varphi_Y(x)\delta(s \mid s, x)\, \mathbb{E}_{G_s}[V_Y] && \text{[Model 1a, averaged over (20)]} \\ \text{not yet treated:} \quad & \lambda_Y(s)\varphi_Y(x)\, \mathbb{E}_{R_s}[V_Y] && \text{[(15)]} \end{aligned}$$

$$
\begin{aligned}
\mathbb{E}_{G_s}[V_Y] &= \frac{\mathbb{E}_{R_s}[V_Y V_S]}{\mathbb{E}_{R_s}[V_S]} && \text{[(20): } G_s = G_{R_s} \text{ reweighted by } v_S\text{]} \\
&= \mathbb{E}_{R_s}[V_Y] + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s}[V_S]} && \text{[}\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y + \mathrm{Cov}(X,Y)\text{]} \\
\frac{\text{just treated}}{\text{not yet treated}} &= \delta(s \mid s, x)\Big(1 + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s}[V_Y]\mathbb{E}_{R_s}[V_S]}\Big) && \text{[divide; } \lambda_Y\varphi_Y \text{ cancel]} \\
&= \delta(s \mid s, x) \cdot \frac{\mathcal{L}\,\mathcal{L}_{ab}}{\mathcal{L}_a\,\mathcal{L}_b}\Big|_{(a_s, b_s)} && \text{[survivor moments from Section 4.3]}
\end{aligned}
$$

- Observed ratio = causal effect × **selection factor**.
- Selection factor = 1 iff V_Y, V_S uncorrelated in the risk set: then timing is as good as random at s.
- The selection factor is known from Step A, so $\delta(s \mid s, x)$ is identified.

8.4 Step D: beyond the moment of treatment (Prop. 3)

→ *Why now:* Step C only gives δ at $t = s$. After s the treated cohort keeps being reshaped by exits, so we need its full survival curve.

For $y > s$, let $g(y, s \mid x)$ be the density of "treated at s and still unemployed at y", and $D := D(y \mid s, x)$:

$$
\begin{aligned}
g(y, s \mid x) &= \mathbb{E}\big[f_S(s \mid x, V)\ \Pr(Y > y \mid S = s, x, V)\big] && \text{[LIE; Assumption 2]} \\
&= \mathbb{E}\big[\lambda_S(s)\varphi_S V_S e^{-b_s V_S} \cdot e^{-\varphi_Y V_Y[\Lambda_Y(s) + D]}\big] && \text{[(3); (2) with (19)]} \\
&= \lambda_S(s)\varphi_S\, \mathbb{E}\big[V_S\, e^{-(a_s + \varphi_Y D)V_Y - b_s V_S}\big] && \text{[linearity; } a_s = \varphi_Y \Lambda_Y(s)\text{]} \\
&= -\lambda_S(s)\varphi_S(x)\, \mathcal{L}_b\big(a_s + \varphi_Y(x)D,\ b_s\big) && \text{[}\mathcal{L}_b = -\mathbb{E}[V_S e^{-aV_Y - bV_S}]\text{]}
\end{aligned}
$$

Divide by the same expression at $y \downarrow s$ (where $D = 0$), as in (18):

$$
\begin{aligned}
P_1(y \mid s, x) &= \frac{\mathcal{L}_b\big(a_s + \varphi_Y(x)D(y \mid s, x),\ b_s\big)}{\mathcal{L}_b(a_s, b_s)} && \text{[the } \lambda_S\varphi_S \text{ cancel]} \\
&= \mathbb{E}_{G_s}\big[e^{-\varphi_Y(x)V_Y D(y|s,x)}\big] && \text{[rewrite with weights (20)]} \qquad (21)
\end{aligned}
$$

- Left side: data. Right side: everything known from Step A, except D.
- Strictly decreasing in $D \Rightarrow$ invert for $D(y \mid s, x)$.
- Then $\delta(y \mid s, x) = \partial_y D(y \mid s, x)/\lambda_Y(y)$ [fundamental theorem of calculus].
- This is the paper's eq. (10), divided by its value at $y \downarrow s$.

Two selection problems, both handled by knowing G

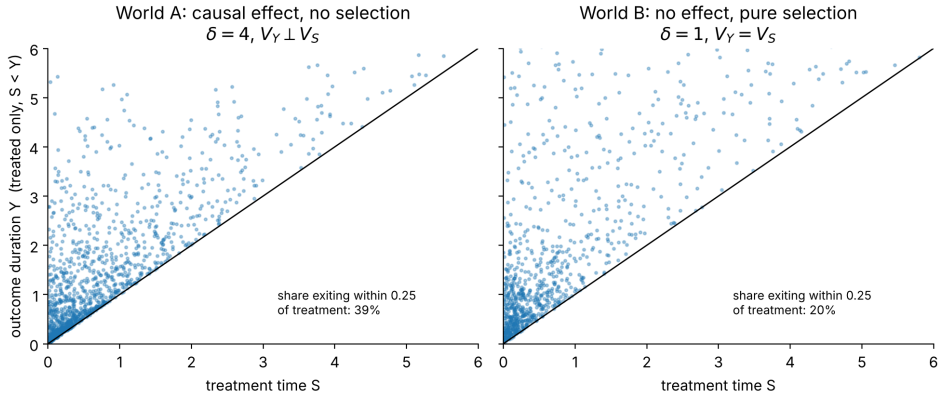
1. Selection *into* treatment at s: the v_S-tilt in (20).
2. Selection *out* after s: in (21), survivors of the treated cohort drift to low V_Y, as in Section 2.
   - Present whenever V_Y varies, even with random timing.
   - So naive hazard ratios at later y are not δ.

8.5 Summary: what is used where

→ *Why now:* collect which assumption did which job.

Assumption	Where it is used
No anticipation	Step A: before S there is no δ, so a clean competing-risks problem
Assumption 2	Steps A, B, D: independence given V; treated = risk set reweighted by v_S
MPH, $V \perp X$	Step A: everything is $\mathcal{L}$ of indices
$\mathbb{E}V < \infty$ (A4)	Step A, step 1: short-duration hazard ratios
Open set (A3)	Step A, step 2: trace $\mathcal{L}$ in 2-D $\Rightarrow$ dependence of V_Y, V_S

- Not used: an excluded instrument, selection on observables, parametric forms, multiple spells.



Intuition. Left: causal effect, no selection; exits bunch right after treatment at every S (local). Right: no effect, pure selection; S and Y correlated overall (global).